



Structure & Syllabus for Semester-VI
BACHELOR OF COMPUTER APPLICATIONS (BCA)
Programme

Gujarat University
2025 - 2026

As per NEP 2020 CURRICULUM AND CREDIT FRAMEWORK FOR
UNDERGRADUATE PROGRAMMES, UGC

&

Resolution No. KCG/admin/2023-24/0607/kh.1

of

Education Department, Govt. of Gujarat

STRUCTURE FOR SEMESTER - 6
GUJARAT UNIVERSITY
Bachelor of Computer Applications (B.C.A.)
(as per NEP 2020)

COURSE:	Bachelor Of Computer Applications (B.C.A.)
MAJOR:	COMPUTER APPLICATIONS
MINOR:	1. COGNITIVE MATHS
	2. DATA SECURITY
	3. WEB TECHNOLOGIES

SEMESTER – 6

MAJOR

CODE	COURSE	CREDITS
DSC-C-BCA-361T	Computer Network	4
DSC-C-BCA-362T	Cloud Computing	4
DSC-C-BCA-363P	System Development Project (External Evaluation of SDP1 & SDP2)	4

MINOR

CODE	COURSE	CREDITS
DSC-M-BCA-364P	Advanced Python Programming	4
	OR	
DSC-M-BCA-364P	Digital Forensic Practical	4
	OR	
DSC-M-BCA-364P	Web Applications Development Framework using Laravel	4

ABILITY ENHANCEMENT COURSE ()

AEC-BCA-365	(Any ONE Course to be selected from the Basket)	2
-------------	---	---

SKILL ENHANCEMENT COURSE
(Any ONE Course. Course can also be chosen from Basket)

SEC-BCA-366	Internship in Major Specific Course	4
-------------	-------------------------------------	---

TOTAL CREDITS		22
----------------------	--	-----------

Semester – VI

BACHELOR OF COMPUTER APPLICATIONS

MINORS

- 1. COGNITIVE MATHEMATICS (CG)**
- 2. DATA SECURITY (DS)**
- 3. WEB TECHNOLOGIES (WTECH)**

Course Name: Web Application Development Framework Using Laravel

Course Code:

Credits: 4

Course Outcomes:

On the completion of the course students will:

- To understand the LARAVEL framework
- To understand how to implement LARAVEL
- To understand how to design and develop responsive web application using LARAVEL

Prerequisites:

HTML, CSS and Basics of PHP

UNIT	TOPIC/SUBTOPIC	TEACHING HOURS	Credits
1	Introduction to LARAVEL	15	01
	Why Use a Framework? A Short History of Web and PHP Frameworks, What's So Special About Laravel?, How It Works, Why Laravel? Setting Up a Laravel Development Environment: System Requirements, Composer, Local Development Environments, Creating a New Laravel Project, Laravel's Directory Structure, Configuration, An Introduction to Artisan Tool, Basic Artisan commands.		
2	LARAVEL Basics	15	01
	Router and Controllers: Route Definitions, Route Groups, Views, Controllers, Route Model Binding, Route Caching, Form Method Spoofing, CSRF Protection, Redirects, Aborting the Request, Custom Responses Blade Template: Echoing Data, Control Structures (Conditionals & Loops), Template Inheritance: @include, @extends, @section, @yield, @stack and @push, View Composers and Service Injection, Custom Blade Directives.		
3	Collecting and Validating User Data	15	01
	Laravel Migration Laravel migration, migration structure, generating migrations, migration commands Laravel file upload, laravel session Laravel database Laravel database, laravel eloquent, laravel relationship model Eloquent orm models: Naming conventions, table name & primary keys, timestamps Basic operations: create, retrieve, update, delete using models, displaying data from models in views		
4	User Authentication and Authorization	15	01

	User authentication and authorization : Using the auth() global helper and the auth façade Validation Defining the routes, creating the controller, writing the validation logic, displaying the validation errors, array validations, creating new validators, error messages & custom errors Available validators: Accepted, after (date), alpha, alpha dash, alpha numeric, array, before (date), between, boolean, date, date format, different, digits, digits between, e-mail, exists(database), image (file), integer, max, min, not in, numeric, regular expression, required, string custom validation rules.		
--	--	--	--

Laravel Project/Practical: To-Do List Web Application

Project Objective:

Design and develop a To-Do List Web Application using the Laravel framework, starting from scratch and implementing features such as adding, displaying, editing, validating, and deleting tasks.

Phase	Project Tasks/Practical Questions
1	Project Setup & Routing Basics (Tasks 1–10) 1. Install Composer on your system and verify using terminal. 2. Create a new Laravel project named `todoapp` using Composer. 3. Run the Laravel project using `php artisan serve`. 4. Explore the default folder structure and understand the MVC architecture. 5. Create a route `/welcome` that returns a simple welcome message. 6. Create a route `/tasks` that returns a hardcoded array of tasks. 7. Create a route `/task/{name}` to return a dynamic task name via URL. 8. Create a route `/task/{name}/{priority}` to accept multiple parameters. 9. Create a Blade view `tasklist.blade.php` to display task names. 10. Use Blade `{{ }}` syntax to display values dynamically in the view.
2	Blade Template Engine & View Management (Tasks 11–20) 11. Pass data from route to view using `with()` method. 12. Use `compact()` to pass multiple task values to the view. 13. Create a master layout `layouts.master.blade.php` using `@yield`, `@section`, and `@extends`. 14. Create reusable header and footer using `@include`. 15. Push custom texts or messages using `@push` and render with `@stack`. 16. Display tasks dynamically using `@foreach`. 17. Use `@if` to check if task is completed. 18. Use `@unless` to show a message when no tasks are present. 19. Use `@switch` to change display based on task status. 20. Create 5 sample tasks using `@for` loop in the view.
3	Forms, CSRF & Validation (Tasks 21–30) 21. Design a form to add a new task (title and description). 22. Use `@csrf` to protect the form from cross-site request forgery. 23. Create a POST route `/tasks/store` to accept the form input. 24. Create a controller `TaskController` and define `store()` method. 25. Validate inputs using `\$request->validate(['title'=>'required'])`. 26. Display validation errors using `@error('title')` and `\$message`. 27. Show all validation errors using `\$errors->all()` in the form view.

	28. Use `withErrors->any()` to display a general validation alert. 29. Use `session()->flash('success')` to show a success message. 30. Redirect back to the form using `return redirect()->back()` after submission.
4	Controllers, Route Management & Session (Tasks 31–40)
	31. Create a `TaskController` using Artisan (`php artisan make:controller TaskController`). 32. Move route logic to controller's `index()` and `store()` methods. 33. Display all tasks from an array in `index()` method. 34. Create a route `/tasks/create` to display form using `create()` method. 35. Use `store()` method to save a new task in session. 36. Create a named route for storing tasks and link to it from form. 37. Use route-model binding (demo with array) to display a specific task. 38. Create a `show(\$id)` method to return a task by index. 39. Protect task creation with dummy middleware (auth simulation). 40. Define resourceful routes using `Route::resource('tasks', TaskController::class)`.
5	Database, Models & Relationships (Tasks 41–50)
	41. Create a migration file for `tasks` table with `title`, `description`, `status`. 42. Set database connection in `.env` file. 43. Run migration using `php artisan migrate`. 44. Create a `Task` model using `php artisan make:model Task`. 45. Update `store()` method to save tasks in database using Eloquent. 46. Update `index()` method to fetch all tasks from DB using `Task::all()`. 47. Create an edit form to update tasks using `edit()` and `update()` methods. 48. Add delete functionality using `destroy()` method and form with `DELETE` method. 49. Add `user_id` column in tasks table and set up foreign key. 50. Establish Eloquent relationships: `User hasMany(Task)` and `Task belongsTo(User)`.

Expected Outcome of this project/practical:

By the end of this case study, students will have:

- Installed and configured Laravel
- Built a working To-Do List app
- Learned Laravel's routing, views, controllers, Blade, validation, sessions, migrations, models, and Eloquent relationships
- Gained practical understanding of building CRUD apps with Laravel

Text Book:

1) Matt Stauffer, "LARAVEL Up and Running, A framework for building modern PHP Apps", 3rd Edition, O'REILLY

Reference Books:

1. Martin Bean, "Laravel 5 Essentials", Packet Publishing
2. Fernando Monteiro, "Hands-On Full-Stack Web Development with Angular 6 and Laravel 5", Packt Publishing
3. Web Technologies : HTML,CSS3, JAVASCRIPT, jQUERY, AJAX, PHP,XML, MVC and LARAVEL), Black Book, 2018, Dreamtech

Web Resources:

- Online Laravel 5.2 Documentation
(<https://laravel.com/docs/5.7>)
- Nathan Wu, Learning Laravel 5 Cookbook
(<https://learninglaravel.net/laravelbook>)
- <https://laravel-news.com/category/laravel-tutorials>
- <https://laravel.com>